Unified Dynamics (Distinction Theory) v2.2: Master Manuscript

1. Front Matter and Abstract

Title: Unified Dynamics: A Typed, Weighted Operator Grammar for Geometry and Matter**Authors:** Benjamin Walker Mayes, et al.**Status:** Final — Canonical Artifact Stack (v2.2)

Abstract

This manuscript formalizes Unified Dynamics (UD) v2.2, a non-perturbative framework modeling physical reality as a typed, weighted operator grammar for geometry and matter. By defining a recursive architecture of distinction and closure on a simplicial substrate, we achieve a synthesis of the 28-prediction scorecard (14 exact, 6 confirmed, 8 candidates). We report the resolution of the cosmological constant gap and the "Null" computational field via the F6 $N=7$ simulation. Results indicate that the injection of simplicial gravity (Einstein coupling $\kappa = 1/9\pi^2$) provides the requisite geometric attraction via curvature defects ($\epsilon = \pi$), creating a potential well that clamps previously monotone computational fields into stable matter packets.

Companion Header

Feature,Geometry Companion (Relaxed Coarse Isotropic Limit),Matter Companion (Minimal Two-Primitive Operator Set)
Primary Claim,General Relativity emerges as the coarse isotropic relaxed limit of the substrate response.,"Particle families are generated by a minimal set of two primitive operator blocks (S_0, H)."
Formalism,"$G_{\mu\nu} = \kappaT^{(GR)} + \Delta T^{(\theta, a, c, b)}$",$Q_{obs} = Q_0(1 + z_Q)$
Logic,Substrate response to burdened states (curvature as transport mismatch).,State grammar defining stable burdened states (matter as persistent defect structure).
Key Equation,Corrected Einstein Block with four accommodation channels.,"Seed-and-Lift ladder (e, μ, τ mass operators)."

2. Part I: Ontological Grammar and Primitive Substrate

The Core Grammar: Simplicial Evolution

The transition from a primary distinction to recursive complexity follows a rigorous nine-stage hierarchy of operator flow:

1. **Point (Primitive Locus):** The foundational possibility of identity; the initial locus of distinction.
2. **Line (First Distinction):** Binary linkage of two points; the act of non-closure differentiation.
3. **Triangle (First Coherent Relation):** The minimal closed loop; the birth of structural coherence.

4. **Hexagon (Surround Path):** Operator flow bifurcation into the "Surround" channel (C_2 wrapping); planar completion.
5. **Pyramid (Lift Gesture):** Flow bifurcation into the "Lift" channel (∂ boundary extension); dimensional escape.
6. **Tetrahedron (First Locked Lock):** Completion of the lift; the first stable 3D volumetric primitive.
7. **Octahedron (First Balanced Compound):** Resolution of polarity; two pyramids joined at the base.
8. **Icosahedron (First Global Shell):** Efficient 3D spherical enclosure; 20 triangular faces.
9. **600-Cell (Recursion):** 4D recursive shell; the previous shell becomes the building block for the next iteration.

Substrate Primitives

The substrate comprises **Locations** and **Transport Links** . The node state is formalized as the triplet $\Psi = (\rho, \theta, \chi)$, where ρ denotes burden (unresolved load), θ denotes local orientation, and χ denotes the specific closure branch.

Binary-Ternary Unification

The substrate is fundamentally ternary, defined by 3 axes with 3 states ($+1, 0, -1$), yielding $3^3 = 27$ configurations.

- **Particle-Aspect (Binary):** Closure-entities require discrete identity. The "neither/between" zero-states are filtered, leaving $2^3 = 8$ binary configurations for observation.
- **Field-Content:** The 19 configurations involving at least one zero-state ($27 - 8 = 19$) constitute the "field content" present across the substrate.
- **Deficit:** The binary deficit $\Delta=7$ governs particle-regime interactions, while the full ternary deficit (26) governs cosmological scaling.

3. Part II: The Retained Existence Engine (REE)

Axioms of Retained Existence

- **X.1 Distinction:** Legibility emerges only at the appearance of a distinction.
- **X.2 Local Binary Retention:** Every distinction induces a local keep/release event $\{0,1\}$.
- **X.3 Recursive Accumulation:** Retained distinctions integrate recursively across the graph.
- **X.4 Mechanical Formation Floor:** Minimum viability requires closure, transport, and burden tolerance.
- **X.5 Triadic Existence Agreement:** Consensus between Omega logic (Ω), Admissibility (Adm), and Support (Supp).
- **X.6 Closure:** Stability is contingent upon locked, non-loose-ended state configurations.
- **X.7 Temporal Retention:** Momentary closure must persist through time to manifest as "world."

- **X.8 Stratified Selection:** Selection is governed by authority layers (Formation $\to$ Retention $\to$ Strain $\to$ Propagation).
- **X.9 Selector Asymmetry:** Foundational gate collapse is terminal; secondary failure induces adaptive strain.
- **X.10 Existence Update:** Structure is continually reselected via the Update Engine (U).

The Construction Spine

The ontological progression is formalized via the Construction Spine: **Distinction $\to$ Binary retention $\to$ Recursive accumulation $\to$ Triadic agreement $\to$ Closure $\to$ Temporal retention $\to$ World.**

Precedence Resolution Order

When overlapping conditions exist, the Update Engine (U) resolves states according to the following hierarchy:| Precedence Order | Resulting Outcome | Description || ------ | ------ | ------ || 1. **Dissolve** | Highest Authority | Total structural erasure. || 2. **Expel** | Terminal | Removal from the world-graph. || 3. **Fragment** | Terminal | Dissolution of coherent identity. || 4. **Repair** | Adaptive | Restoration via drift sensitivity. || 5. **Strained** | Adaptive | Survival under secondary gate failure. || 6. **Propagate** | Extension | Expansion of coherent structure. || 7. **Stable** | Ground State | Maintenance of the current state. |

4. Part III: Universal Structure Theorem and Simplicial Variation

Simplicial Boundary Channels

The K_4 (Tetrahedron) structure supports independent variation channels through its graded simplex boundary operator (∂) and co-boundary (δ):

- **δ^0 (Vertices $\to$ Edges):** Maps to **Frame Variation** .
- **δ^1 (Edges $\to$ Faces):** Maps to **Connection Variation** .
- **δ^2 (Faces $\to$ Body):** Maps to **Curvature Integration** .

4 π Angular Normalization Lemma

The dimensionless coupling $\alpha_{body} = 1/\pi^2$ is derived by normalizing the K_4 coupling against the full angular support of the local 3D orientation field:

1. Let $g_{body}^2 = 4/\pi$ (derived from the Regge curvature quantum $\epsilon = \pi$).
2. Define $\Omega = 4\pi$ as the complete angular support of a unit sphere in 3D.
3. The dimensionless coupling is obtained by normalizing the squared coupling: **$\alpha_{body} = g_{body}^2 / \Omega = (4/\pi) / (4\pi) = 1/\pi^2$** .

5. Part IV: The Regime-Conditional Framework

The Master Formula and RG Ladder

Discrepancies between K_4 predictions ($v = 246 \text{ GeV}$) and observation are resolved via the Regime-Conditional Formula: $$P_{obs}(\mu) = P_{K4}(v) \times C_{QCD}(\mu/v) \times C_{EW}(\mu/v) \times C_{grav}(\mu/M_{Pl})$$ The **IFS inverse levels**

(n) serve as the Renormalization Group (RG) ladder, where energy decades and threshold crossings (Λ_{QCD}) occur between discrete ticks of the inverse IFS.

K_4 Beta Function

The rate of QCD running is determined by the distinction number $\Delta = 7$: **$b_0 = \frac{33 - 12}{12\pi} = \frac{21}{12\pi} = \frac{7}{4\pi} = \frac{\Delta}{4\pi}$** .

Regime Classification

Class,Observables,Status
A: Direct,"α_s , n_s , m_W , Leptons",Validated ($C \approx 1$)
B: Small Regime,"θ_{12} , m_H , λ",Validated ($C \approx 1 \pm 10\%$)
C: Large Regime,"Quarks, Neutrinos, δ_{CP}",Open (Requires explicit RG)

6. Part V: Geometry, Simplicial Gravity, and F6 Results

Modified Gravitational Hamiltonian

Simplicial gravity is injected into the OCPF Hamiltonian to account for curvature defects in the δ^1 channel: $$H_{grav} = -JK + \lambda V + \kappa \cdot \hat{R}*{K_4}$$ *where* ***$\kappa = 1/9\pi^2$*** *and* *$\hat{R}*{K_4}$* is the Regge deficit angle operator.

Inflection and CC Resolution

In the $N=7$ monotone (null) run, the violation term (V) acted as pure repulsion. Injecting κ introduces **Geometric Attraction** . The curvature defect $\epsilon = \pi$ creates a potential well pulling states toward the K_4 fixed point.

- **Inflection Point:** $\lambda/J \approx 88.4$.
- **Phase Shift:** f_{close} probability jumps from 0.34 to 0.92 .
- Gravity serves as the "Temporal Glue" necessary to hold information together once the $N=7$ partition threshold is exceeded.

7. Part VI: The Unified Lepton Grammar

Seed-and-Lift Operators

- **$S_0 = 43/7$ (Neutral-Seed Threshold):** Six coherence slots plus a $1/7$ irreducible remainder (formation memory).
- **$H = 101/6$ (Universal Lift):** 16-node inherited scaffold plus 1 burden crossing, minus a $1/6$ coherence reserve.

The Charged-Lepton Ladder

- **Electron (e):** $\nu_0 S_0$
- **Muon (μ):** $e \cdot S \cdot H$ (where $S = 2S_0$)
- **Tau (τ):** $e \cdot S \cdot H^2$| Ratio | UD Bare Form | Value | Observed | Residual (z) || ------ | ------ | ------ | ------ | ------ || m_μ / m_e | $S \cdot H$ | 207.21 | 206.77 | $\approx -2 \times 10^{-4}$ || m_τ / m_μ | H | 16.833 | 16.82 |

$\approx -8 \times 10^{-4}$ || m_τ / m_e | $S \cdot H^2$ | 3490.37 | 3477.15 | $\approx -1.5 \times 10^{-3}$ |

8. Part VII: Yin/Yang Regime Structure and Symmetry

Reality is defined by two interpenetrating sublattices of the diamond lattice:

- **$+1$ Regime (Matter):** Positive action-cost, closure-entities, particle projection.
- **-1 Regime (Geometry):** Mirror sector; sign-flipped operations are admissible.
- **Symmetry:** This dual-sublattice structure provides the basis for CPT Symmetry and the Dirac sea.
- **Charge:** $q \in \{-1, 0, +1\}$ is a direct projection of these regime sectors.

9. Part VIII: The 28-Prediction Scorecard

Quantity,Exact Form,Value,Status,Meaning
Master Scalar x,$7/192$,0.03646,Exact,Survival trace: $(1/2) \cdot (7/16) \cdot (1/6)$
Spectral Index n_s,$185/192$,0.96354,Exact Bare,Bare CMB spectral tilt.
Tensor Ratio r,$7/192$,0.03646,Bare Spine (Falsified),Primordial GW amplitude (Exceeds bound <0.030).
Koide Ratio Q_K,$2/3$,0.66667,Exact,Lepton triplet structural ratio.
Fine Structure α_0^{-1},$960/7$,137.142,Candidate,Interaction intensity (x/5).
$N=7$ Inflection,$\lambda/J \approx 88.4$,—,Confirmed,Matter-packet formation point.

10. Part IX: Resolution of Gaps 1–7

- **Gap 1:** Global closure via Banach fixed-point convergence.
- **Gap 2 (Red Tilt):** Derived from Ω -convergence. As Ω converges to 1, power is imprinted on larger scales.
- **Gap 3 (Kinetic Term):** Derived from the **Genesis Action Cost** $\hbar_d$. Making a distinction is not free. In the slow-genesis limit, the absolute-value action $|\dot{a}|$ of the Poisson process reduces to the quadratic form $\frac{1}{2}\dot{a}^2$.
- **Gap 5 (Fractal Dimension):** $d_H \approx 2.965$ confirmed via Box-Counting, Pair Correlation (d_2), and Mass-Radius. The 2D measurement (1.965) is corrected by the embedding factor ($d_H(3D) = d_H(2D) + 1$).| Feature | Inflation | UD (Ω -Convergence) || ------ | ------ | ------ || **Mathematical Basis** | Potential flatness | Banach Fixed-Point Theorem || **Tensor Ratio** | 16ϵ | ≈ 0 (IFS contraction) |

11. Part X: Conclusion and Artifact Registry

The Master Rule

$Q_{obs} = Q_0(1 + z_Q)$ Reality is the weighted image of a typed operator grammar, where exact spines (Q_0) are dressed by native accommodations (z_Q).

Artifact Index and Governance

1. **Foundational Grammar:** Seed $\to$ threshold $\to$ lift $\to$ dressing.

2. **Corrected Einstein Block:** $G_{\mu\nu} = \kappaT_{GR} + \Delta T_{\theta, a, c, b}$.
3. **Operator Core:** $S_0 = 43/7, H = 101/6$.
4. **Edit Propagation:** Revisions to lower-tier artifacts (S_0) necessitate recalculation of all dependent lepton and oscillation artifacts. The $N=7$ inflection and $7/192$ ratio are locked invariants.